E-Commerce Customer Shopping Behavior Analysis

Submitted By
Aditya Raj
Bachelor of Technology (B.Tech.) in Computer Science and Engineering
(Third Year)

Technical Stack:
Microsoft Excel • Python (Jupyter Notebook) •PostgreSQL • Tableau

CONTENTS

S.No.	Chapter Title
1	Abstract
2	Acknowledgment
3	Executive Summary
4	Problem Statement
5	Tools & Technologies Used
6	Dataset Description
7	Methodology & Dashboard
8	Conclusion
7	Appendix Code

📄 ABSTRACT

This project, “E-Commerce Customer Shopping Behavior Analysis”, focuses on analyzing transactional data to uncover meaningful business insights and support data-driven decision making. The primary objective of this study is to identify revenue patterns, high-value customer segments, category performance, and geographical sales contribution.

The dataset consists of 700 records and 12 variables, including invoice details, product categories, pricing, quantity, and country information. Using **SQL for data cleaning and feature engineering**, **Python (Pandas, NumPy, Matplotlib) for validation and outlier detection using the IQR method**, and **Tableau for visualization and dashboard creation**, the raw data was transformed into structured business intelligence.

Key findings include:

- Total Revenue: **$22.4K**
- Total Orders: **549**
- Average Order Value (AOV): **$40.77**
- Kitchen & Garden categories dominate revenue share
- Australia contributes the highest revenue (~$6.3K)
- Hong Kong shows the highest AOV (~$237)

ACKNOWLEDGMENT

I would like to express my sincere gratitude for the opportunity to work on this self-initiated project, *E-Commerce Customer Shopping Behavior Analysis*.

This project was independently conceptualized, designed, and executed as part of my continuous learning journey in the field of Data Analytics. Throughout the development process, I relied on academic knowledge, online documentation, and open-source learning resources to strengthen my understanding of SQL, Python, and Tableau.

I would like to acknowledge the Kaggle platform for providing the dataset that enabled practical, real-world analysis. The availability of open-source tools and community resources played a significant role in enhancing the quality of this project.

This project has significantly improved my practical skills in data cleaning, exploratory data analysis, visualization, and business interpretation, and has strengthened my confidence in solving real-world analytical problems independently.

Executive Summary

This project focuses on analyzing customer purchasing behavior and revenue patterns to identify high-value segments and optimize business strategy.

Using Python (Pandas, NumPy, Seaborn ,Matplotlib) SQL and Tableau, raw transactional data was cleaned, transformed, and analyzed to uncover actionable insights.

Key findings revealed revenue concentration among top customer segments, category-wise sales contribution, and seasonal growth trends.

The analysis provides data-driven recommendations that can improve marketing targeting, inventory planning, and overall revenue growth.

Problem Statement

In today's competitive e-commerce environment, businesses generate massive transactional data but often fail to convert it into actionable insights.

Companies struggle to:

i. identify high-value customers
ii. Understand category performance
iii. Optimize revenue growth strategies
iv. The decision-making gap arises due to the lack of structured data analysis and effective data visualization.
v. This project aims to bridge that gap by applying systematic data analytics techniques to generate meaningful and actionable insights.

❖ Tools & Technologies Used

- Python (Pandas, NumPy, Matplotlib, Seaborn)
- SQL (PostgreSQL)
- Tableau
- Excel
- Basic Statistics

❖ Dataset Description

Source: Kaggle

Dataset Size:

Number of Rows: 700

Number of Columns: 12

Numerical Variables:
invoice no, unit price , order value, quantity

Categorical Variables:
country, invoice time, year month, major category, minor category, description

Date-Time Variable:
invoice_date

Methodology

Step 1: Data Cleaning (Using SQL)
Checked total records using COUNT()
Verified duplicate entries using GROUP BY and HAVING clause
Confirmed no NULL values were present in the dataset
Validated data types and column consistency

Step 2: Data Validation
Verified logical relationship: Revenue = unit_price × quantity
Ensured proper date-time formatting for invoice_date
Outlier detection using IQR method (Python – Jupyter Notebook)

Step 3: Feature Engineering (Using SQL)
Added a new revenue column for business performance measurement
Generated aggregated revenue insights using SUM() and GROUP BY.

Step 4: Exploratory Data Analysis (EDA)

Revenue Distribution
Analyzed overall revenue spread to identify high and low sales ranges.
Detected revenue concentration patterns and potential outliers.
Assessed revenue variability to understand business performance stability.

Category-Wise Sales Analysis
Evaluated total revenue generated by each major category.

Identified top-performing and underperforming product segments.
Compared category contributions to overall sales.

Country-Wise Contribution
Analyzed revenue contribution across different countries.
Identified key international markets driving business growth.
Supported geographical expansion strategy insights.

Customer Segmentation
Segmented customers based on purchase behavior and revenue contribution.
Identified high-value and frequent buyers.
Enabled targeted marketing and retention strategies.

Step 5: Visualization
Line Charts for Monthly Growth
Bar Charts for Category Analysis
Pie Charts for Customer Segments

Step 6: Dashboard Creation
Interactive dashboard created using Tableau for business-level reporting.

Business Impact

The analysis enabled identification of high-performing categories (Kitchen & Garden), allowing businesses to prioritize inventory and marketing efforts for maximum revenue generation.
Country-level insights highlighted Australia, France, and the UK as key revenue drivers.
Detection of high Average Order Value in Hong Kong indicates premium product positioning opportunities.
Monthly revenue trend analysis revealed seasonal fluctuations and volatility.
Product-level analysis showed that a small group of products contributes significantly to total revenue.
MoM growth analysis helped identify periods of strong growth and decline.

🔎 Revenue & Growth Dynamics

The business generated $22.4K in total revenue across 549 transactions, with an Average Order Value (AOV) of $40.77.
Revenue trends show cyclical volatility with peak months exceeding $2.5K.
Month-over-Month growth exceeding 300% in certain periods.

🏷 Category-Level Performance

Revenue concentration is highest in the Kitchen and Garden categories.
Revenue distribution suggests a semi-Pareto pattern.

🌍 Geographic Insights

Australia (~$6.3K) is the primary revenue contributor.
Hong Kong shows the highest Average Order Value (~$237).

📦 Product-Level Concentration

Top-selling SKUs disproportionately contribute to total revenue.

☑ Strategic Interpretation

Revenue variability suggests opportunities for demand forecasting enhancement.

SQL & Python Implementation Screenshots

Query Query History

```sql
SELECT invoice_no, unit_price, quantity, order_value, country,
       invoice_date, invoice_time, year_month,
       major_category, minor_category, description,
       COUNT(*) AS duplicate_count
FROM ecom_clean
GROUP BY invoice_no, unit_price, quantity, order_value, country,
         invoice_date, invoice_time, year_month,
         major_category, minor_category, description
HAVING COUNT(*) > 1;
```

Data Output Messages Notifications

invoice_no bigint	unit_price double precision	quantity bigint	order_value double precision	country text	invoice_date date	invoice_time time without time zone	year_month text	major_category text	minor_category text	description text	duplica bigint

Query Query History

```sql
SELECT column_name, data_type
FROM information_schema.columns
WHERE table_name = 'ecom_clean';
```

Data Output Messages Notifications

Showing rows: 1 to 12 Page No: 1 of 1

	column_name name	data_type character varying
1	revenue	numeric
2	unit_price	double precision
3	order_value	double precision
4	quantity	bigint
5	invoice_date	date
6	invoice_time	time without time zone
7	invoice_no	bigint
8	year_month	text
9	major_category	text
10	country	text
11	minor_category	text
12	description	text

description text	revenue numeric (10,2)
SPOTTY BUNTING	29.70
POSTAGE	7.80
RETRO MOD TRAY	90.00
JUMBO BAG DOLLY GIRL DESIGN	19.80
RED RETROSPOT PUDDING BOWL	122.40
PACK OF 20 SPACEBOY NAPKINS	12.50
DOTCOM POSTAGE	66.00
WRAP ENGLISH ROSE	27.20
WOOD STAMP SET THANK YOU	20.00
GREEN POLKADOT BOWL	13.20
TOY TIDY SPACEBOY	29.70
RED RETROSPOT CHILDRENS UMBRELLA	27.20
PACK OF 60 SPACEBOY CAKE CASES	20.00
PACK OF 20 NAPKINS PANTRY DESIGN	10.20
NATURAL SLATE HEART CHALKBOARD	35.40

Query Query History

```sql
SELECT invoice_no, revenue
FROM ecom_clean;
```

Data Output Messages Notifications

	invoice_no bigint	revenue numeric (10,2)
1	554839	29.70
2	549316	7.80
3	541975	90.00
4	541975	19.80
5	564087	122.40
6	549738	12.50
7	564087	66.00
8	553148	27.20
9	564087	20.00
10	553148	13.20
11	564087	29.70
12	553148	27.20
13	564087	20.00

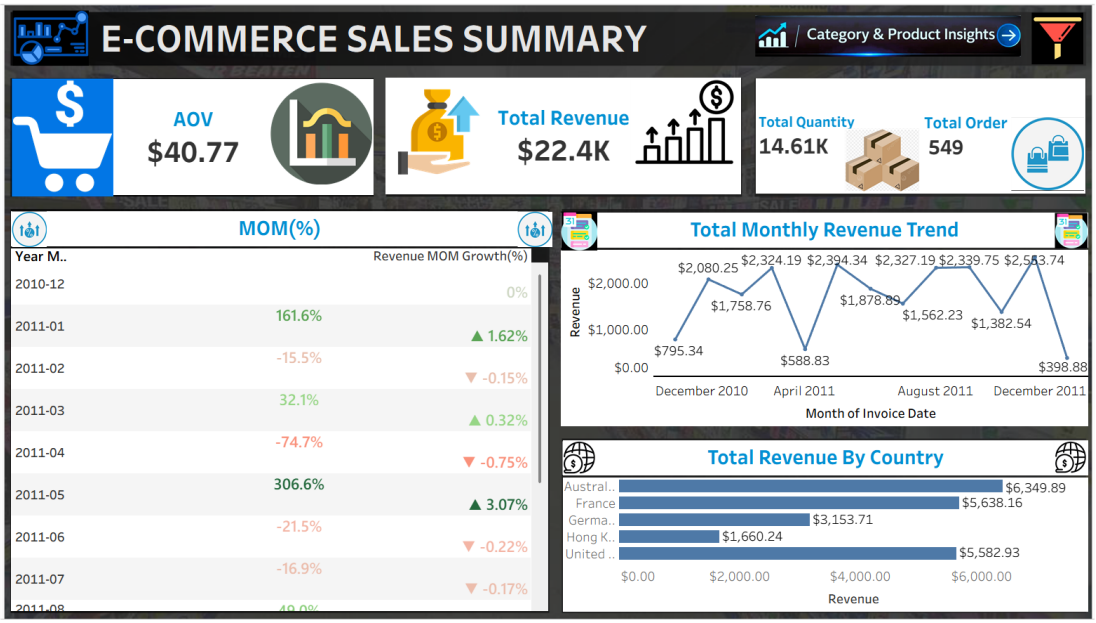
Detecting Outliers using IQR Method

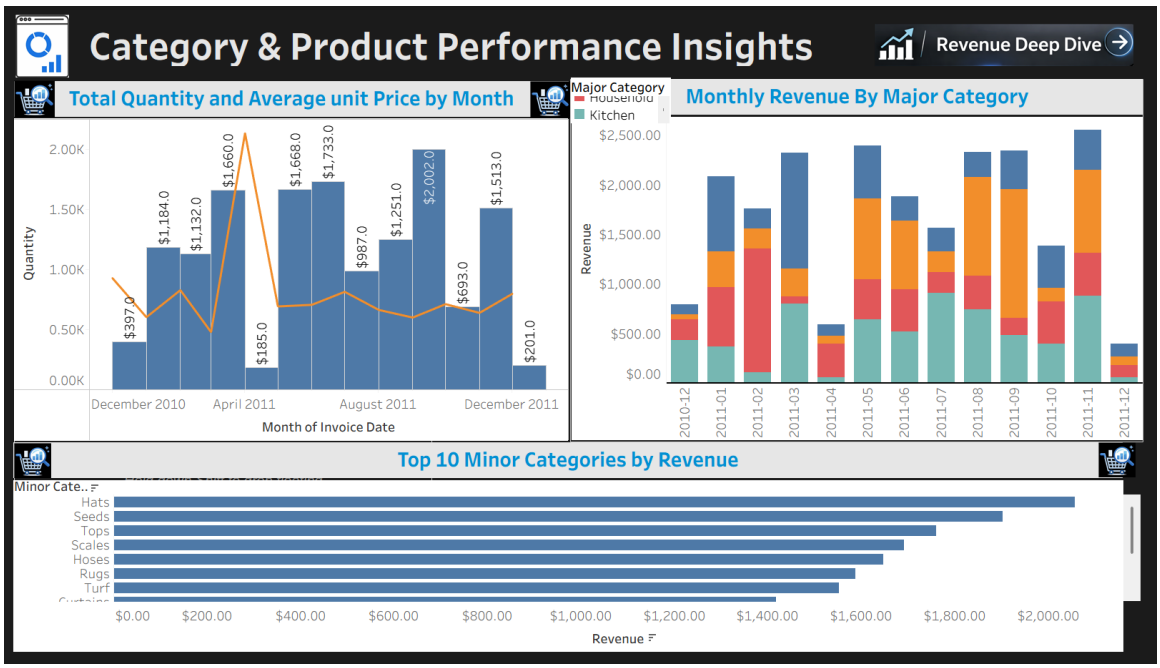
[13]:
```python
Q1 = df['order_value'].quantile(0.25)
Q3 = df['order_value'].quantile(0.75)

IQR = Q3 - Q1

lower_limit = Q1 - 1.5 * IQR
upper_limit = Q3 + 1.5 * IQR

lower_limit, upper_limit
```

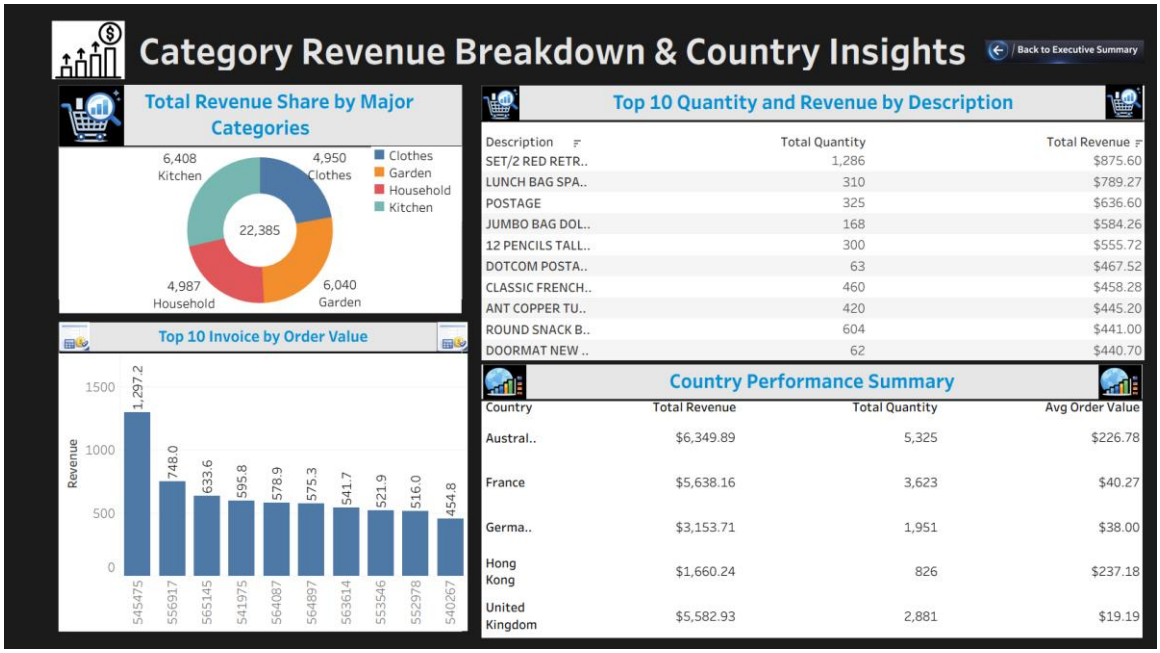
[13]: (-17.183750000000003, 50.74625)

[14]:
```python
df_no_outliers = df[
    (df['order_value'] >= lower_limit) &
    (df['order_value'] <= upper_limit)
]
```

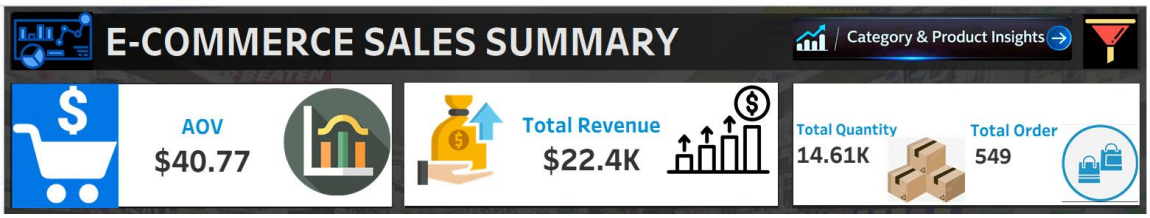
Tableau Dashboard Screenshots







Key Performance Indicator dashboard (KPI)



Conclusion

This project successfully transformed raw transactional data into meaningful business insights using SQL, Python (IQR validation), and Tableau visualization.

The project demonstrates how structured data analytics can convert operational data into actionable business intelligence.

- ## Analytical Insights:-

Total Revenue: $22.4K
Transactions: 549
AOV: $40.77
Kitchen & Garden dominate revenue share.
Australia (~$6.3K) is highest revenue contributor.
Hong Kong shows highest AOV (~$237).

❖ Appendix – Code Implementation

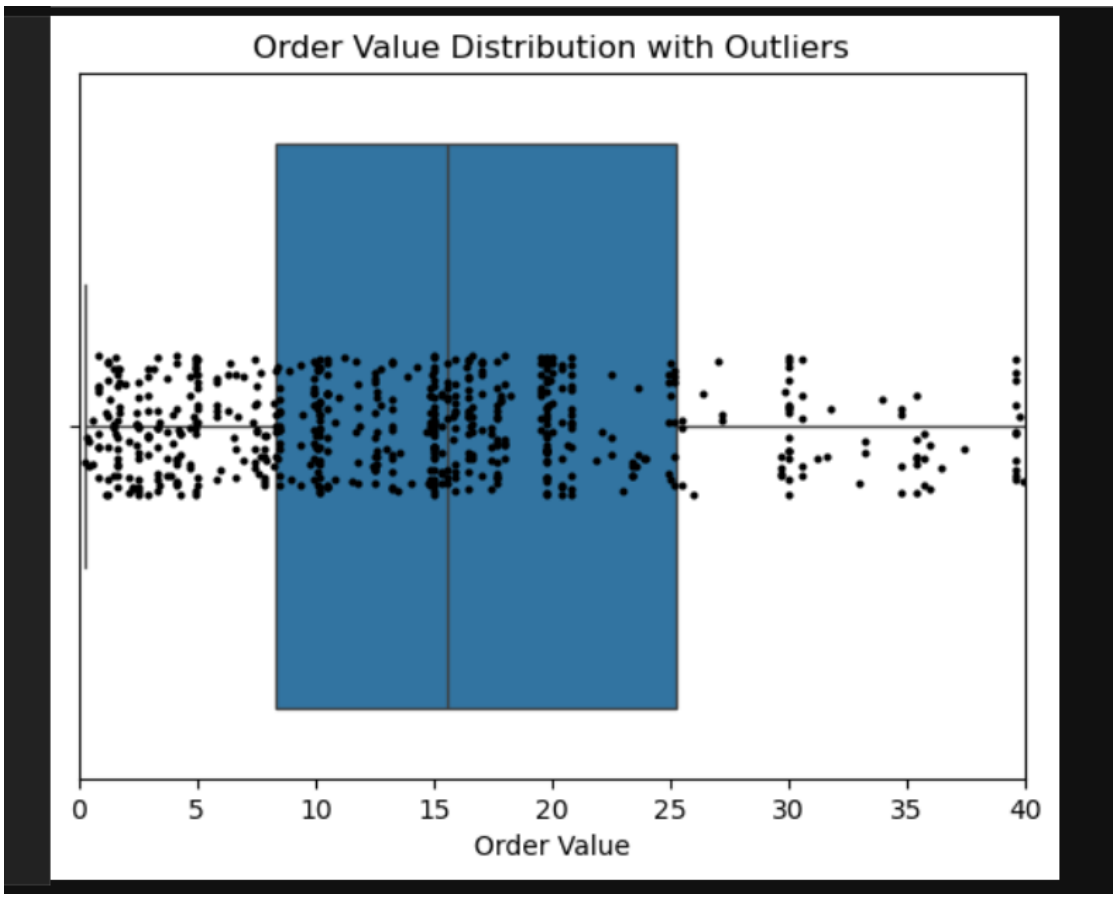
Python – IQR Outlier Detection

Detecting Outliers using IQR Method

[13]:
```
Q1 = df['order_value'].quantile(0.25)
Q3 = df['order_value'].quantile(0.75)

IQR = Q3 - Q1

lower_limit = Q1 - 1.5 * IQR
upper_limit = Q3 + 1.5 * IQR

lower_limit, upper_limit
```

[13]:
```
(-17.183750000000003, 50.74625)
```

[14]:
```
df_no_outliers = df[
    (df['order_value'] >= lower_limit) &
    (df['order_value'] <= upper_limit)
]
```



check how many outlier

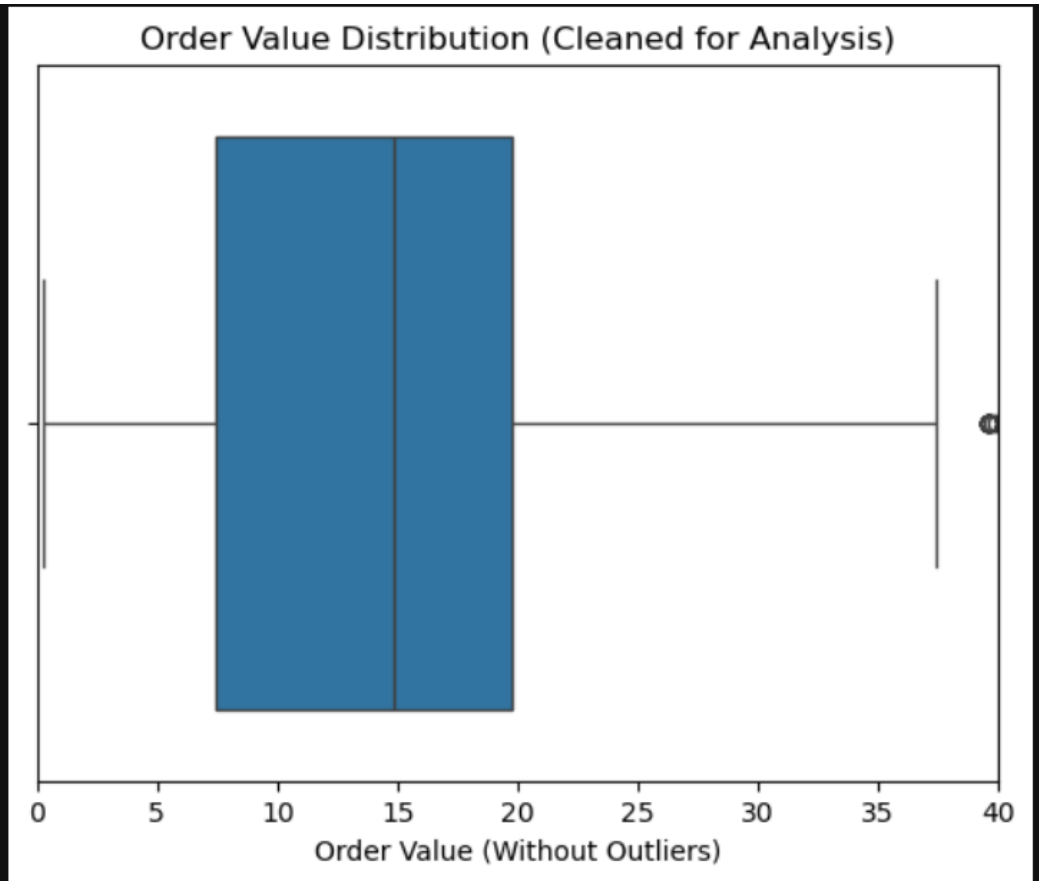
[15]:
```
len(df) - len(df_no_outliers)
```

[15]: 91

Visualizing cleaned order value distribution

after removing outliers using IQR method.

[16]:
```
sns.boxplot(x=df_no_outliers['order_value'])
plt.xlabel("Order Value (Without Outliers)")
plt.xlim(0, 40)
plt.title("Order Value Distribution (Cleaned for Analysis)")
plt.show()
```



References

Data Source

Kaggle. (n.d.). *E-Commerce Dataset*. Retrieved from https://www.kaggle.com/

Programming & Libraries

Python Software Foundation. (n.d.). *Python Documentation*. Retrieved from https://docs.python.org/

Harris, C. R., et al. (n.d.). *NumPy Documentation*. Retrieved from https://numpy.org/doc/

McKinney, W. (n.d.). *Pandas Documentation*. Retrieved from https://pandas.pydata.org/docs/

Waskom, M. (n.d.). *Seaborn Documentation*. Retrieved from https://seaborn.pydata.org/

Matplotlib Development Team. (n.d.). *Matplotlib Documentation*. Retrieved from https://matplotlib.org/stable/contents.html

---

Database & Query Language

PostgreSQL Global Development Group. (n.d.). *PostgreSQL Documentation*. Retrieved from https://www.postgresql.org/docs/

ISO/IEC. (n.d.). *Structured Query Language (SQL) Overview*.

Visualization Tool

Tableau Software. (n.d.). *Tableau Documentation*. Retrieved from https://www.tableau.com/support

Statistical Concepts

Montgomery, D. C., & Runger, G. C. (n.d.). *Applied Statistics and Probability for Engineers*.

Project Resources:-

Complete Project Folder :

Ecommerce_Sales_Analytics_Project

Thank you for your time and consideration